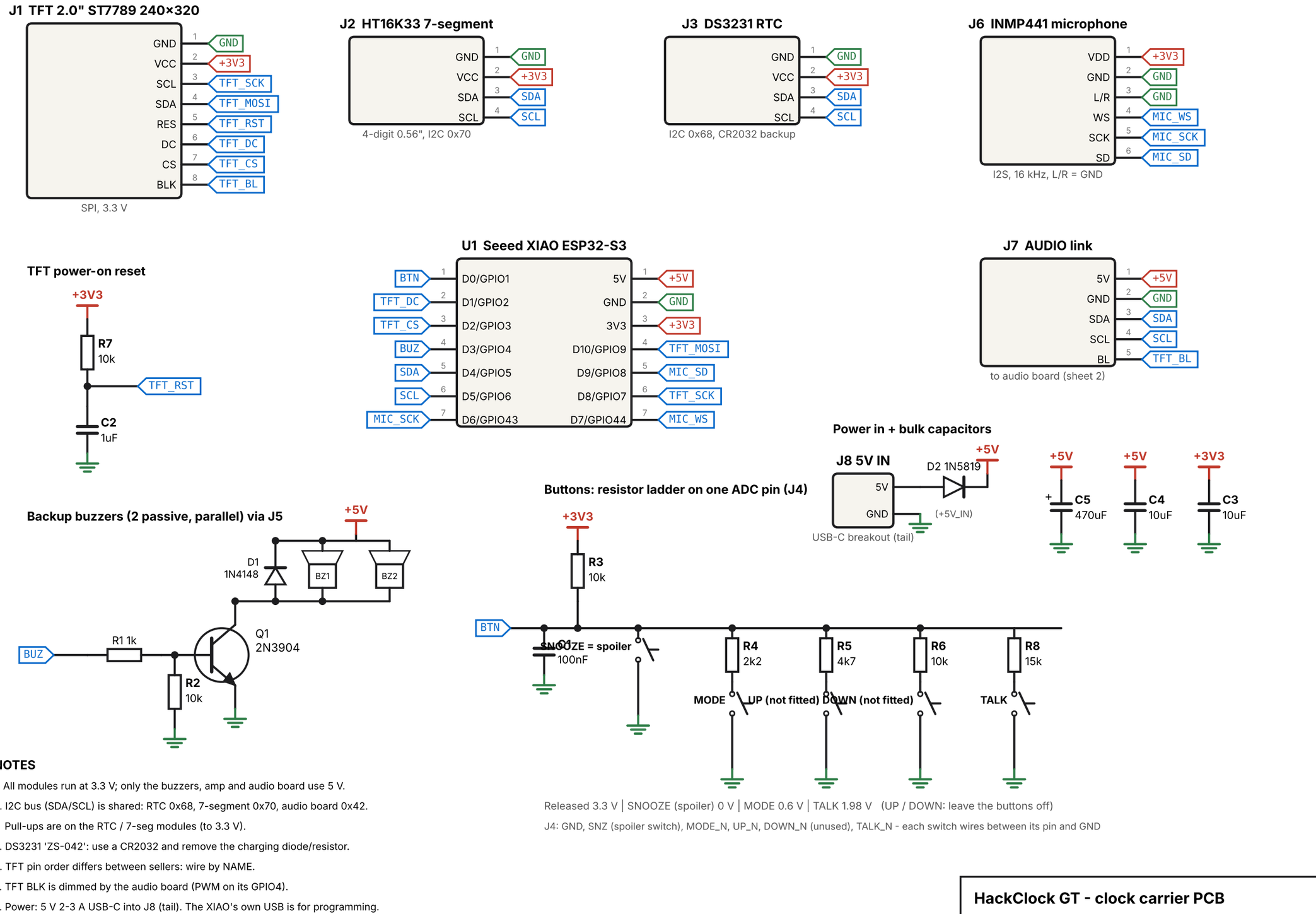


NOTES

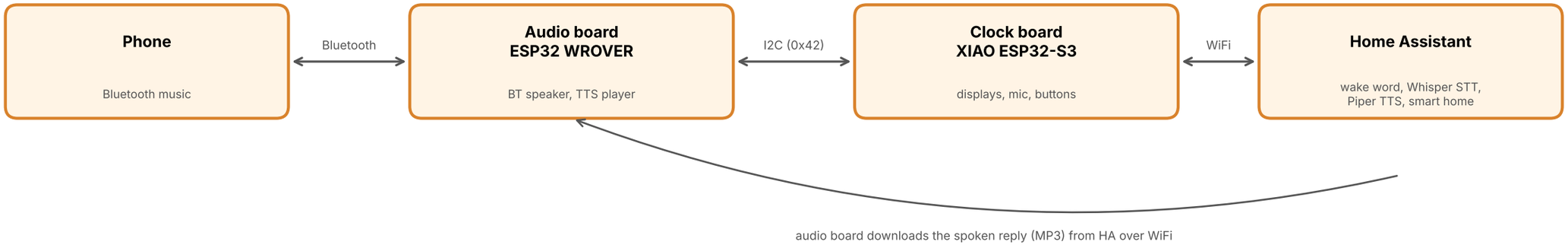
1. All modules run at 3.3 V; only the buzzers, amp and audio board use 5 V.
2. I2C bus (SDA/SCL) is shared: RTC 0x68, 7-segment 0x70, audio board 0x42.
   Pull-ups are on the RTC / 7-seg modules (to 3.3 V).
3. DS3231 'ZS-042': use a CR2032 and remove the charging diode/resistor.
4. TFT pin order differs between sellers: wire by NAME.
5. TFT BLK is dimmed by the audio board (PWM on its GPIO4).
6. Power: 5 V 2-3 A USB-C into J8 (tail). The XIAO's own USB is for programming.

HackClock GT - clock carrier PCB

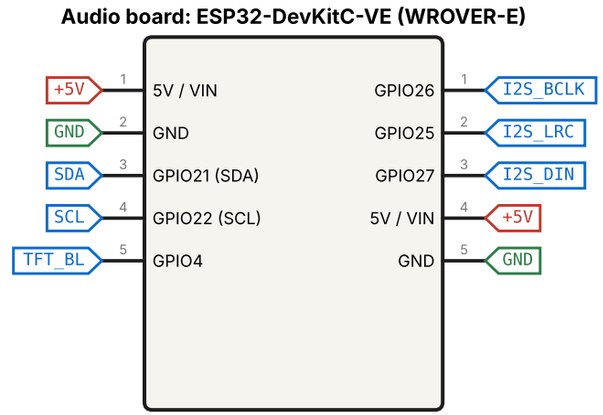
Hack Club Half Life | rev 3.2 | sheet 1/2

Same names = same net (net-label schematic)

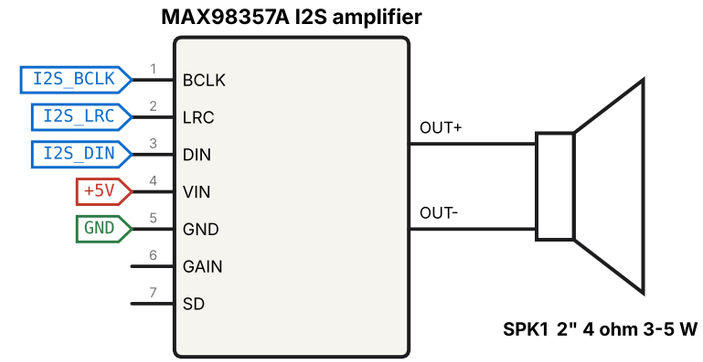
System overview



to clock PCB J7 "AUDIO"



any classic ESP32 with PSRAM - NOT an S3/C3 (no Bluetooth Classic)



3 W class-D, mono

GAIN, SD: leave open (9 dB gain, L+R mix)

Audio board notes

- I2S_* nets are short Dupont wires from the ESP32 board straight to the amp (not on the PCB).
- Speaker wires: solder to the amp's screw terminal / pads; keep them twisted.
- Both boards run from the 5 V USB-C input (J8). The amp draws ~0.8 A at full volume.
- Flash the audio board once over its own USB port (firmware/audio-board), then it runs on its own.
- Phone pairing name: "HackClock". The clock board shows track title/artist on its "Now playing" page.
- WiFi details are sent to the audio board by the clock over I2C - nothing to configure.

Power + charging



Samsung watch puck
middle of the roof, own adapter

The watch puck is NOT wired to the PCB: its cable runs inside the body and out through the USB-A slot in the tail to its own adapter (or a multi-port charger).

HackClock GT - audio, power + system

Hack Club Half Life | rev 3.2 | sheet 2/2

Same names = same net (net-label schematic)